

Release from PI: An analysis and a model

D. J. K. Mewhort¹ · Kevin D. Shabahang¹ · D. R. J. Franklin¹

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Abstract Recall decreases across a series of subspan immediate-recall trials but rebounds if the semantic category of the words is changed, an example of release from proactive interference (RPI). The size of the rebound depends on the semantic categories used and ranges from 0% to 95%. We used a corpus of novels to create vectors representing the meaning of about 40,000 words using the BEAGLE algorithm. The distance between categories and spread within categories jointly predicted the size of the RPI. We used a holographic model for recall equipped with a lexicon of BEAGLE vectors representing the meaning of words. The model captured RPI using a hologram as an interface to bridge information from episodic and semantic memory; it is the first account of RPI to capture release at the level of individual words in categorized lists.

Keywords interference · holographic memory · semantic memory · dynamic storage

In the Brown–Peterson paradigm, subjects are given a series of trials each composed of a short (subspan) list of words, and the subjects are required to recall the words after studying each list. If the words fall in one semantic category, the number of words recalled correctly will decrease across the series of trials, an example of proactive interference (PI). If the semantic category is changed following a series of trials from a

single category, however, performance will rebound; the rebound is called release from PI.

Figure 1 illustrates the phenomenon.¹ In the experiment, separate groups of subjects received four trials for immediate recall; each trial involved three words. In the control condition, all the words were from a single category (fruits). In the experimental conditions, the category of words was switched on the fourth trial from the first category to the second (professions to fruits, meats to fruits, and so forth).

As shown in Fig. 1, in both the experimental and control conditions, performance dropped across the first three trials from about 90% correct to about 30%. On the fourth trial, performance in the control condition continued to decrease; in the experimental conditions—the switched-category conditions—performance rebounded. The extent of the rebound depended on the categories involved.

In such experiments, intrusions (report of words that were not studied on a particular trial) are usually words studied on previous trials. Intrusions that were not taken from previously studied trials are almost always related semantically to the studied list (Hasher, Goggin, & Riley, 1973; Goggin & Riley, 1974).

Wickens (1970, Fig. 7) examined the size of the release for 13 pairs of category shifts and showed that it depends on the specific categories (release from about 95% to near zero); he argued the size of release reflects overlap of features when the words are encoded. In terms of the data illustrated in Fig. 1, he argued:

This paper originally published with an incorrect figure display. The paper has been corrected.

✉ D. J. K. Mewhort
mewhortd@queensu.ca

¹ Department of Psychology, Queen’s University, Kingston, Ontario, Canada

¹ We used Xyscan to obtain numerical values for the data from his figure. The software can be downloaded from <http://rhig.physics.yale.edu/~ullrich/software/xyscan/>

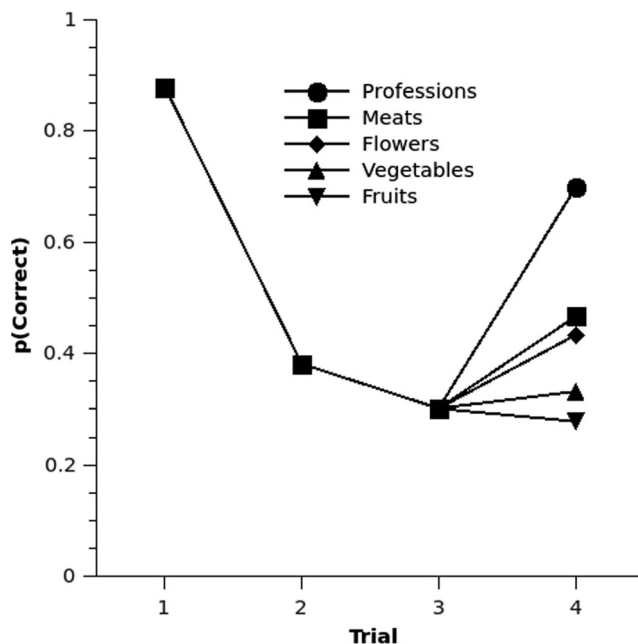


Fig. 1 Accuracy of report as a function of practice and change of category (from professions to fruits, meats to fruits, flowers to fruits, vegetables to fruits) on the final trial. The figure was redrawn from Wickens (1973, Fig. 4)

Fruits and vegetables might be encoded by means of two common attributes. They are things to be eaten and things grown from the ground. The meats are appropriate to the first attribute (they are edible) but not to the second (they do not grow from the ground). Flowers, however, grow from the ground but are not usually edible. Names of professions share neither attribute. (Wickens, 1973, p. 489)

Wickens's (1973) account depends on the match between the features encoded during the induction trials and the features encoded during the release trial. The account is based on his view of how encoding works, a view that was quite radical for the time. Briefly, the then-dominant position treated words as undefined entities and held that encoding meant transferring them to a labile short-term store—also known as primary memory (Atkinson & Shiffrin, 1968; Waugh & Norman, 1965). By contrast, Wickens suggested that words are represented by a set of descriptive features and that both the features and encoding reflect the meaning of the word. When we encode a word, it is filtered into an appropriate hierarchy of categories as it is entered into short-term memory:

“Horse” might be encoded as beasts of burden, four-legged creatures, mammals, warm-blooded animals, and finally of animals in general. In short, I suspect that the encoding process functions in the manner of a good player of Twenty Questions, but in more or less the reverse direction. (Wickens, 1970, p. 1)

Wickens's (1973) discussion of the encoding and classification—the idea that fruits and vegetables might be encoded as things to be eaten and as things grown from the ground—was limited by the fact that he did not have a computationally tractable way to represent the meaning of words and objects. Aside from pioneering work on the semantic differential (Osgood, 1952; Osgood, Suci, & Tannenbaum, 1957), there was little contemporary work on how a word's meaning might be specified and, in particular, how it might be built into a computable representation. More recent work has developed ways to represent the meaning of words that give us practical methods with which to build the necessary representation of meaning (e.g., M. N. Jones & Mewhort, 2007; Landauer & Dumais, 1997; see also Mandera, Keuleers, & Brysbaert, 2016; Pereira, Gershman, Ritter, & Botvinick, 2016).

The present work explores the use of semantic information derived using the BEAGLE algorithm, a procedure for building a representation of meaning developed by M. N. Jones and Mewhort (2007; see also M. N. Jones, Kintsch, & Mewhort, 2006). BEAGLE vectors represent the meaning of a word in terms of its position in a hyperspace. The BEAGLE algorithm constructs vectors defining the hyperspace by recording how each word is used in each sentence of a large corpus of text.

BEAGLE's representation of meaning

BEAGLE builds a representation of the meaning of a word by constructing a history of its use in text (see M. N. Jones & Mewhort, 2007; M. N. Jones, Willits, & Dennis, 2015). Two words are similar to the extent that they share the same pattern of usage in their history. For instance, the words *cat* and *dog* are similar because they take up similar roles when they occur within sentences (e.g., the sentences “the _ is on the mat” and “my lovely pet _” could be completed by either *cat* or *dog*). Their common history of usage determines their similarity in meaning.

BEAGLE learns two kinds of information from text. The first is *item information*, a record of which other words are used in the same sentence as the target word; the information is stored in item vectors, one vector for each unique word in the text. The second is *order information*, a record of where the target word is placed in the sentence; the order information is stored in order vectors, one for each unique word in the text. Finally, the context and order vectors are combined to yield lexical vectors for each unique word in the text.

BEAGLE assigns a placeholder to each unique word in the text. The placeholders are vectors of values sampled at random from a Gaussian probability distribution; the placeholders are called visual vectors, v_i . BEAGLE uses the placeholders to construct each word's history of usage in the text on a sentence-by-sentence basis. If a sentence in the text

were “Cats like mice,” the item information taken from the sentence about the word *cats* is the sum of the visual vectors assigned to the other words in the sentence. If $\mathbf{v}_{like}$ and $\mathbf{v}_{mice}$ are visual vectors for the words *like* and *mice*, the item vector for *cats*, $\mathbf{m}_{cats}$, would be updated by the sum of the random vectors for the other two words in the sentence:

$$\mathbf{m}_{cats} = \mathbf{m}_{cats} + \mathbf{v}_{like} + \mathbf{v}_{mice}$$

From only one sentence, the item vector for *cats* would not change very much, but if the representation is summed across several thousand sentences involving *cats*, the item representation converges to a stable history of *cats* as it is used in the corpus. In effect, each sentence contributes a modest amount of information about how the word has been used, and the noise inherent in the initial visual vectors is averaged out.

The order information taken from the sentences is constructed by summing vectors that summarize where the target word appears in the sentence. First, the position of the word is coded using a placeholder vector (a fixed Gaussian vector). Then, order information is formed by binding the position of the target word with all n -grams in the sentence that include the word. If the placeholder vector is designated Φ , there are three vectors that locate the word *like* in the sentence, namely, $(\mathbf{v}_{cats} \otimes \Phi)$, $(\Phi \otimes \mathbf{v}_{mice})$ and $(\mathbf{v}_{cats} \otimes \Phi \otimes \mathbf{v}_{mice})$, where $\otimes$ denotes a binding operator.

The binding operator is a form of directional circular convolution (see M. N. Jones & Mewhort, 2007; Kelly, Blostein, & Mewhort, 2013; Plate, 2003). Order vectors are created using the same summation idea that is used to construct the item vectors, but, instead of summing visual vectors representing individual words, the bound convolutions representing the order information are summed. In the example, the order information for *like* from the sentence *Cats like mice* would be $(\mathbf{v}_{cats} \otimes \Phi) + (\Phi \otimes \mathbf{v}_{mice}) + (\mathbf{v}_{cats} \otimes \Phi \otimes \mathbf{v}_{mice})$.

The representation of meaning in BEAGLE depends on both the corpus of text and the set of visual vectors. Suppose, for example, that you were to construct a set of vectors twice using the same text corpus for both sets. If the initial Gaussian visual vectors are different for the two sets, vectors for the corresponding words will not share the same elements. Although the corresponding vectors will not share elements, the vectors will define the same similarity structure, measured using cosines; the relation amongst words within a given set will be similar across sets.

A further note of caution is in order: The meaning can be unexpected. One might expect, for example, that colour names would always fall into the category *colour*, but in a corpus based on old novels, *violet* may fall into a category of proper names as well as *colour*. *Violet* is a colour, of course, but word usage sometimes yields surprising results depending on the text.

Clearly, the vectors produced using the BEAGLE algorithm depend critically on the corpus of text on which they are based. Johns, Jones, and Mewhort (2016) have improved the performance of language-based cognitive models by optimizing the choice of corpora used to fit benchmark tasks. They argued that optimizing the choice of corpora, in effect, optimizes a parameter often neglected in fitting cognitive models: the linguistic experience that the subject brings to language-based tasks.

BEAGLE and release from PI

A BEAGLE vector represents the meaning of a word as a position in a hyperspace. Our first question is whether such a representation is relevant to release from PI. As Wickens, Dalezman, and Eggemeier (1976) note,

the problem, of course is to find a system which would predict semantic differences before beginning the experiment. A potential solution to this problem is to make a prediction based upon the amount of overlap of denotative attributes between groups and to vary the amount of overlap among groups predicting varying amounts of release from PI as a function of the amount of overlap. (p. 307)

In the spirit of their suggestion, we created BEAGLE vectors by reading a text corpus composed of novels (see M. N. Jones & Mewhort, 2007, for details of how to construct BEAGLE vectors). The corpus contained 39,076 unique words in 10,238,600 sentences. To build the BEAGLE vectors, the dimensionality was set to 1,024, a value large enough to provide good resolution without excessive strain on our computer resources.

Next, we turned to Wickens's (1970) documentation of the size of the release in terms of the relation between the inducing and the release categories. We took the words that Wickens (1970) used and analyzed the corresponding item, order, and lexical vectors. The appendix lists the source of the materials, the materials and Wickens's abbreviation for the pairs of stimuli in parentheses.² Because BEAGLE vectors do not currently contain information about the physical attributes of the stimuli (such as font, colour, or modality of presentation), we were unable to use his data on the size of release based on those attributes.

² Where possible, we used the vectors corresponding to the same words that Wickens (1970) had used in each of the experiments that he reported. Sometimes, however, the words were not listed in the publication, but the criteria used to select the words were reported. In such cases, we based our selection of words on the criteria.

We analyzed the item and order vectors for the target words separately. First, we calculated the extent to which the vectors within a category clustered. To do so, we calculated the centre of each category of words (the mean of its vectors) and then calculated each word's Euclidean distance from the centre of its category. Secondly, we calculated the Euclidean distance between each word in one category and the centre of the other category.

The first measure (spread) is a way of defining how tightly words within a given category clump together. It was used because our intuition is that categories that are tightly clustered around their centre should have a greater impact than those that are diffusely represented in the hyperspace. The second measure (separability) defines the distance of one category from the centre of the other category. Our intuition is that categories that are widely separated will provide greater contrast when used in a release from PI paradigm.

Table 1 shows the mean spread and distance measures for each class of words, along with the release from PI, for the experiments of interest. The table also shows the correlation between Release from PI and each predictor separately (bottom row).

To summarize the relation between release from PI and the spread and distance measures, we calculated an overlap score that combined the two measures for each pair of categories:

$$\text{Overlap}(\mathbf{x}, \mathbf{y}) = \frac{\text{Spread}(\mathbf{x})\text{Spread}(\mathbf{y})}{\text{Separability}(\mathbf{x}) + \text{Separability}(\mathbf{y})}$$

$$= \frac{\sum_1^m \text{Distance}(\mathbf{x}_j, \mu_x) + \text{Distance}(\mathbf{y}_j, \mu_y)}{\sum_1^m \text{Distance}(\mathbf{x}_j, \mu_y) + \text{Distance}(\mathbf{y}_j, \mu_x)},$$

Table 1 Spread, separability, and the size of the release from PI

Case	Spread		Separability		RPI
	Item	Order	Item	Order	
NP	0.81	0.79	0.75	0.72	0.19
NS	0.80	0.69	0.79	0.68	0.20
AA	0.84	0.69	0.81	0.67	0.25
Fr	0.87	0.72	0.77	0.66	0.50
WN	0.96	1.02	0.55	0.51	0.94
TC	0.97	0.85	0.71	0.59	0.73
SI	0.89	0.76	0.78	0.62	0.57
D_E	0.72	0.78	0.64	0.73	0.56
D_A	0.74	0.81	0.66	0.74	0.63
D_P	0.81	0.78	0.62	0.64	0.72
Im	0.95	0.88	0.81	0.71	0.14
VA	0.79	0.96	0.75	0.73	0.01
<i>r</i>	0.77	0.70	0.25	0.20	

where $\mathbf{x}$ and $\mathbf{y}$ are matrices for the two classes with m different word vectors. *Distance* refers to Euclidean distance. The centre of the j th category is represented by μ_j . *Overlap* ranges from zero to one, where overlap of 1 means complete overlap.

To illustrate how overlap works, Fig. 2 presents the position of words in a compressed semantic space. The axes are the three principal components of the feature space derived from the 39,076 word vectors. The top panel shows the position of words for a pair of low-overlap categories (*fruits* and *professions*, overlap = 0.73); the bottom panel shows the position of words for a pair of high-overlap categories (*fruits* and *flowers*, overlap = .84). As shown, words in the low-overlap pair are well separated while those in the high-overlap pair intermingle in the space.

Figure 3 presents the relation between the size of the release (release from PI taken from Wickens, 1970) and the overlap measure. The top panel shows the relation using BEAGLE's item vectors and the bottom panel shows the relation using BEAGLE's order vectors. As is clear in the figure, overlap is strongly correlated with Wickens's (1970) release measure, although it is stronger using item vectors ($r = -.79$) than order vectors ($r = -.51$). We conclude, therefore, that BEAGLE vectors are a good predictor of release from PI. The prediction works because the vectors capture the overlap in meaning between categories.

Tulving (1972, 1985) distinguishes two memory systems: *Semantic* memory holds general knowledge (e.g., that *apple* is a *fruit* or how to tie your shoelaces), whereas *episodic* memory contains information bound to a specific time and place (e.g., that you studied *apple* on a particular list). Unfortunately, aside from labeling the interaction between the two systems (top-down processing), little is known about how semantic information affects episodic memory. The release from PI paradigm presents an interesting challenge because it forces us to address how meaning affects ongoing episodic processing. In the following, we develop a mechanism to handle the challenge illustrated by Wickens's (1970, 1973) examples. Our mechanism serves, in effect, as an interface between the two memory systems.

A process model

Figure 3 shows that structural similarity among BEAGLE vectors predicts the size of Release from PI. But prediction is only half the battle. We need to understand the mechanism by which meaning affects study and recall to drive performance in the release from PI paradigm.

Franklin and Mewhort (2002, 2013, 2015) offer a possible mechanism based on holographic storage (see Gabor, 1969). They start from the premise that subjects in a memory

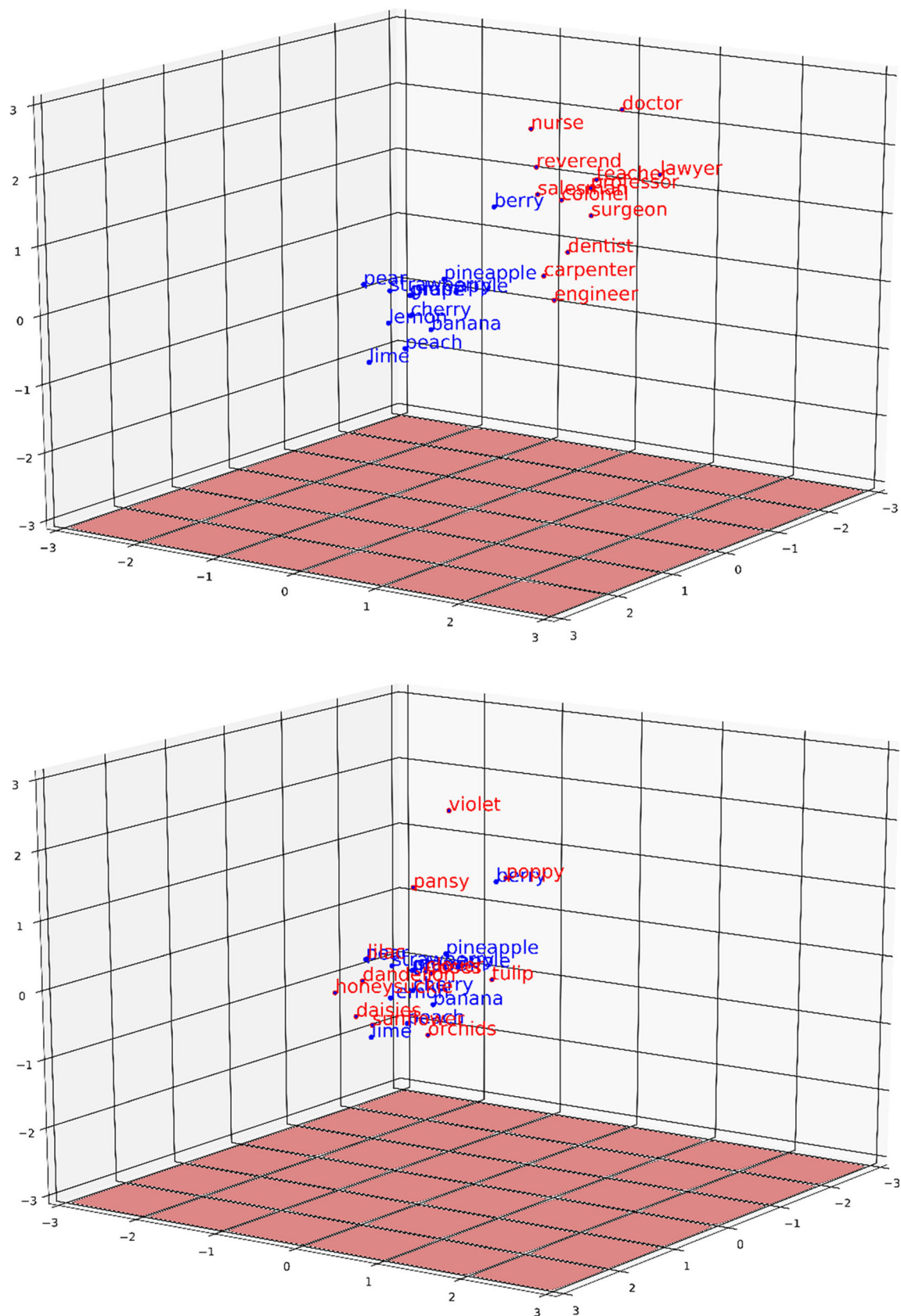


Fig. 2 Overlap between *fruits* and *professions* (top panel) and between *fruits* and *flowers* (bottom panel). (Colour figure online)

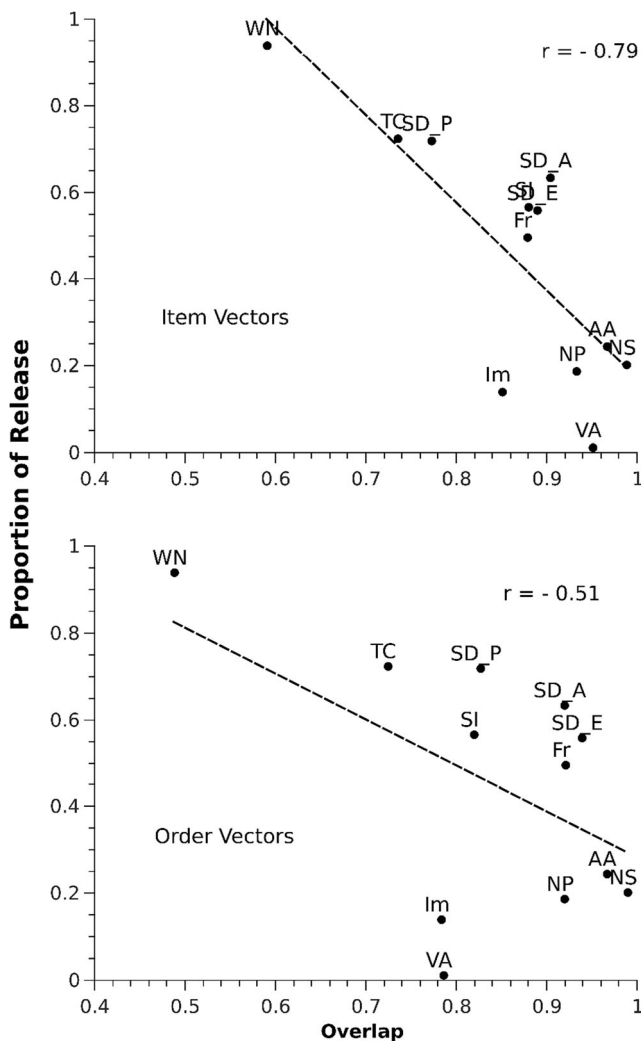


Fig. 3 Overlap between categories and the size of the release from PI (RPI). RPI scores were taken from Wickens (1973). *Top panel* presents the data using BEAGLE's item vectors; *bottom panel* presents the corresponding data using BEAGLE's order vectors for the same words used by Wickens

experiment already know the words used as stimuli. Hence, studying words in the experiment does not teach the words or transfer them from an external buffer to an internal representation. Rather, the hologram serves as the subject's working lexicon; studying a list reinforces or strengthens the words in the hologram.

Franklin and Mewhort (2015) emphasize three convenient properties of a hologram (see also Plate, 2003, pp. 140–142). Firstly, a hologram is address-free; one can recover an item from a hologram without implying that the item has an address. Secondly, a hologram is robust to loss of medium. One can, for example, destroy a large amount of the hologram without catastrophic loss of the information stored. Third, a hologram is a dynamic data structure. Adding an item to the hologram changes the strength of all items in the hologram in

proportion to their similarity to the items stored in it. The change can be both positive and negative.

To represent items in the subject's lexicon, Franklin and Mewhort (2013, 2015) used Gaussian vectors centred at zero with a variance of $1/2048$, (one over the dimensionality of the vectors; cf. TODAM; Murdock 1982, 1983). To simulate a list-learning experiment, all words and word-to-word associations were preloaded into the model's lexicon at a resting level. The preloaded material was intended to represent the subject's prior knowledge.

When studying a list of words, the model associated successive words, using one-way circular convolution. Next, it added the item to the lexicon (weighted to reflect the item's position in the input stream) and added the newly created association to the hologram. In practice, the links form a chain of interitem associations or, perhaps, a set of fragmented chains.

To recall, the model combines information about each potential response from two sources: information about the item in the hologram and information contributed by associations to the item in the hologram. The model reports the item with combined strength closest to a criterion.

Adapting the model to use BEAGLE vectors. To apply the holographic model to release from PI, we used BEAGLE vectors instead of the Gaussian vectors that Franklin and Mewhort (2013, 2015) had used. Specifically, we used the BEAGLE vectors generated from a corpus of novels (the same vectors used to develop Fig. 3).

Gaussian vectors are well behaved. They are, for example, orthonormal in expectation. In other words, they are approximately independent of each other and have an expected length of 1.0. Moreover, if one builds a similarity matrix for Gaussian vectors with each other (using the dot product to measure similarity), the distribution of similarities is symmetrical. BEAGLE vectors are less well behaved. They are certainly not independent of each other—if they were, they would not exhibit the properties shown in Fig. 3 and would be useless as predictors of word meaning and release from PI.

Figure 4 shows the distribution of similarities (measured using a dot product) of BEAGLE's item vectors with each other (i.e., the values below the principle diagonal of a similarity matrix) along with the corresponding distribution for Gaussian vectors. As is shown in the top panel of the figure, the distribution for the Gaussian values is symmetrical around zero. As shown in the bottom panel, the distribution for the BEAGLE vectors is both shifted and skewed to the right and has a much larger variance than the corresponding Gaussian distribution. Because the similarity distribution of the Gaussian vectors is centred at zero and is symmetrical, additions to the lexicon are as likely to inhibit other items as to strengthen them.

Because the distribution for BEAGLE vectors is both skewed and shifted, adding an item is more likely to

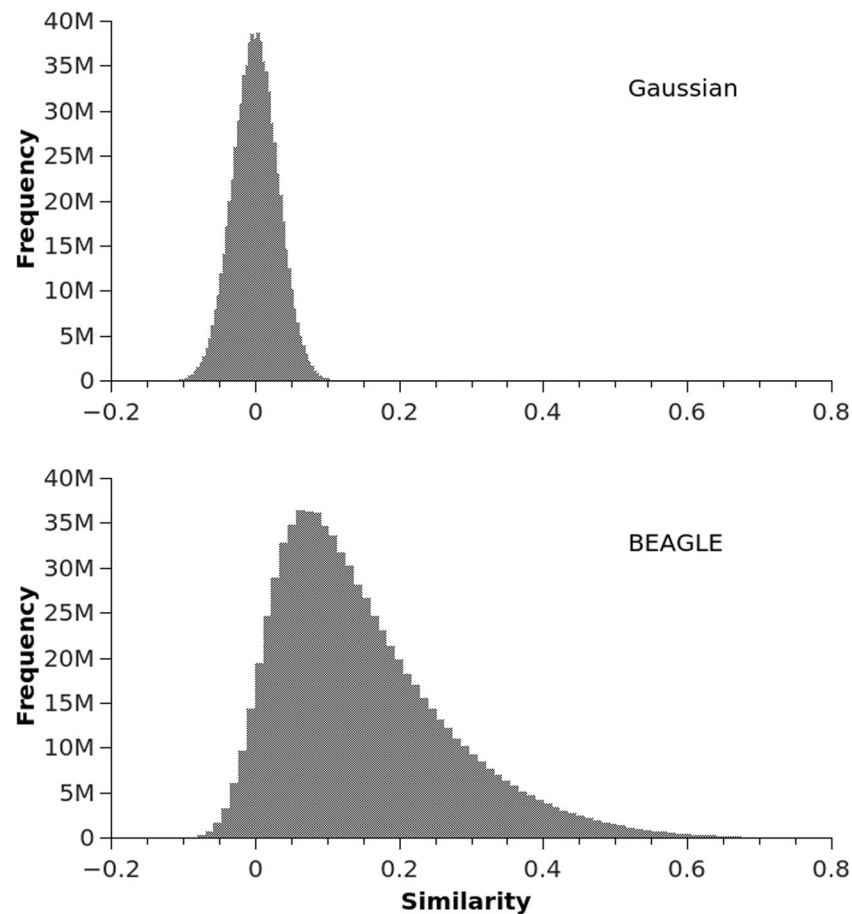


Fig. 4 Frequency of similarity values. *Bottom panel* shows the frequency distribution for all pairs of the 39,076 word vectors used in the simulations. *Top panel* shows the corresponding data for the same number of Gaussian vectors

strengthen than to inhibit other items. Figure 5 provides an example. The figure shows the amount of change to all items in the lexicon as a function of their similarity to a new item when it is added to the hologram. The top panel shows the case using Gaussian vectors, and the bottom panel shows the corresponding data for BEAGLE vectors. Notice that the top distribution is symmetrical around zero whereas the bottom distribution is shifted in the positive direction. Hence, with BEAGLE vectors, the mean strength of items in the hologram will increase as items are added; that is, the hologram's entropy increases uncontrollably. Our observation resonates with warnings about the properties of Gaussian and BEAGLE vectors in formal models of memory (Johns & Jones, 2010).

Because of the differences between BEAGLE and Gaussian vectors, we altered Franklin and Mewhort's (2015) holographic model to hold activation between bounds (rather than allowing it to explode). During the study, when we added an item to the hologram, we weighted it by its distance from the criterion so that items far from the criterion were pushed further than items close to the criterion. We treated both the item information and its accompanying associative information in the same way. The model reported the item closest to

criterion, provided that it was within a specified range; otherwise, we halted. Finally, after each report, we provided feedback based on the report's item information; unlike the study case, we pushed the reported item to the lower bound. Because the release from PI paradigm involves very short lists, we ignored serial position during study. Specifically, we used weights for associative, α , and item, β , information that did not vary across serial position. Finally, we introduced a third parameter, γ , to weigh the feedback.

In formal terms, studying the i th word in the list, $\mathbf{w}_i$, updates the hologram, $\mathbf{L}$, by

$$\mathbf{L} = \mathbf{L} + \alpha \times (\mathbf{C} - s_{\text{associative}}) \times (\mathbf{w}_{i-1} \otimes \mathbf{w}_i) + \beta \times (\mathbf{C} - s_{\text{item}}) \times \mathbf{w}_i,$$

where, $\mathbf{C}$ is the criterion, $\otimes$ indicates directional convolution, $s_{\text{associative}}$ and s_{item} refer to the current associative and item strengths, respectively, and α and β are parameters. The current strengths, $s_{\text{associative}}$ and s_{item} , are calculated as,

$$s_{\text{associative}} = (\mathbf{w}_{i-1} \otimes \mathbf{w}_i) \cdot \mathbf{L} \\ s_{\text{item}} = \mathbf{w}_i \cdot \mathbf{L}.$$

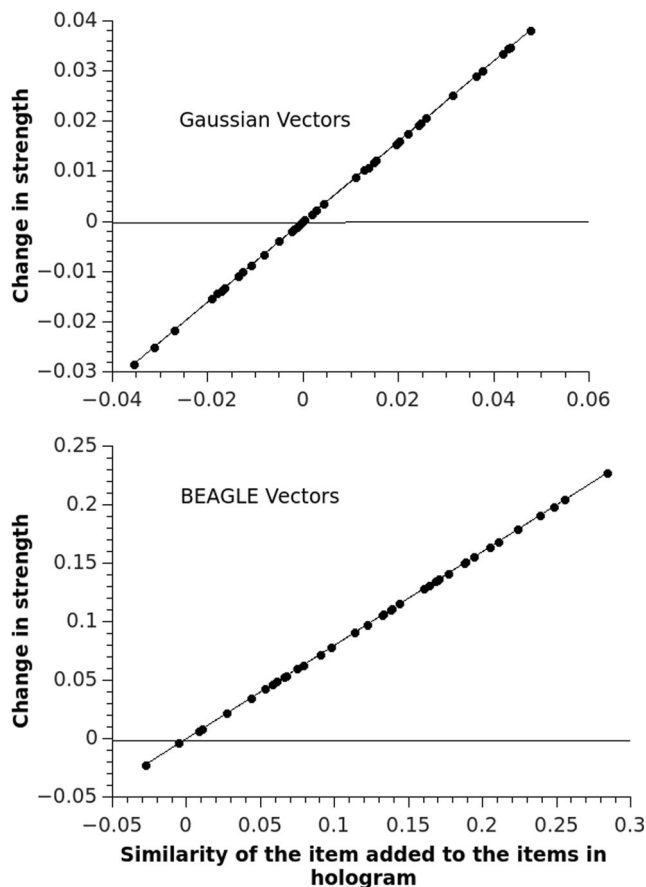


Fig. 5 Changes in the strength of items stored in the hologram as a function of their similarity to an item added to the hologram

Feedback changed the state of the hologram after each item report (either a correct item or an error):

$$\mathbf{L} = \mathbf{L} - \gamma \times (\alpha \times (\mathbf{p} \otimes \mathbf{r}) + \beta \times \mathbf{r}),$$

where $\mathbf{p}$ and $\mathbf{r}$ are the probe and report vectors, respectively, and γ weighs the amount of feedback. At the beginning of report, we used *start* as the instruction to begin studying and to begin report.

Figure 6 shows strength of activation in the hologram for a demonstration task from a single pseudosubject at different steps in a hypothetical experiment. Figure 6 is organized in four panels. We ordered the words in the hologram so that the items used in the experiment would appear in the panel. Before studying any words, activation of all words in the hologram would hover around 0.0. Trial 1, shows the activation of words in the hologram after studying *engineer*, *lawyer*, and *salesman*. Notice that the activation of other words has been lifted off the floor with a greater increase for words in the same category (professions). Trial 2 shows the state of the hologram after studying *professor*, *doctor*, and *colonel*. As before, activation of studied items—and unstudied words

from the same category—has been increased. Trial 3 shows the situation after studying *teacher*, *dentist*, and *surgeon*. Finally, Trial 4 shows the situation after studying *peach*, *strawberry*, and *apple*, words from a new category (fruits). Notice that activation of both the studied items and the members of the fruit category has been raised. Importantly, activation of previously studied words has been suppressed.

The changes in activation as a function of category result from an interaction between the structure of similarity embodied in the BEAGLE vectors and the dynamic nature of storage in a hologram. Figure 6 illustrates the basic mechanism by which the current model obtains release from PI. The pattern of activation and suppression is analogous to the pattern often attributed to spreading activation. In the case of the hologram, however, the changes are automatic and instantaneous: They reflect the structure of the hologram.

Semantic contrast effects

Fitting release from PI To fit the model to existing data, we formed pseudosubjects. Each pseudosubject had a unique combination of words in their lexicon, acknowledging intersubject differences in vocabulary. Averaging across the pseudosubjects produced smooth curves that made it easier to fit the model to experimental data. Each pseudosubject was given a vocabulary (lexicon) of 10,000 words. The vocabulary included the words to be used in the simulated experiment plus a random selection of other words from the 39,076 words available and 100,000,000 forward one-way associations and 100,000,000 backward one-way associations. The lexicon was normalized and weighted by 0.0001. To fit the model to experimental data, we used a SIMPLEX optimization routine to minimize the difference between the model's output and the data (see Press, Teukolsky, Vetterling, & Flannery, 1992, pp. 423–435).

Figure 7 shows the fit to Goggin and Wickens (1973) data. In their control condition, all words were from a single category (body parts or foodstuffs); in the experimental condition, the category used in the control condition was switched to the other (release) category (from body parts to foodstuffs or vice versa). We fit the control condition and then applied the parameters from the fit to the experimental condition. Hence, we did not change the parameters after we had fit the control condition; we merely changed the stimuli presented to the model. Because we did not change the parameters after fitting the control condition, the release from PI shown in the figure is a direct effect of the change in category. As shown in Fig. 7, the model does a good job of capturing Release from PI.

Table 2 shows the parameters used in fitting the model to the simulations shown in Fig. 7 (and in subsequent simulations). The three parameters were denoted α , β , and γ , for weights applied to associative, item, and feedback,

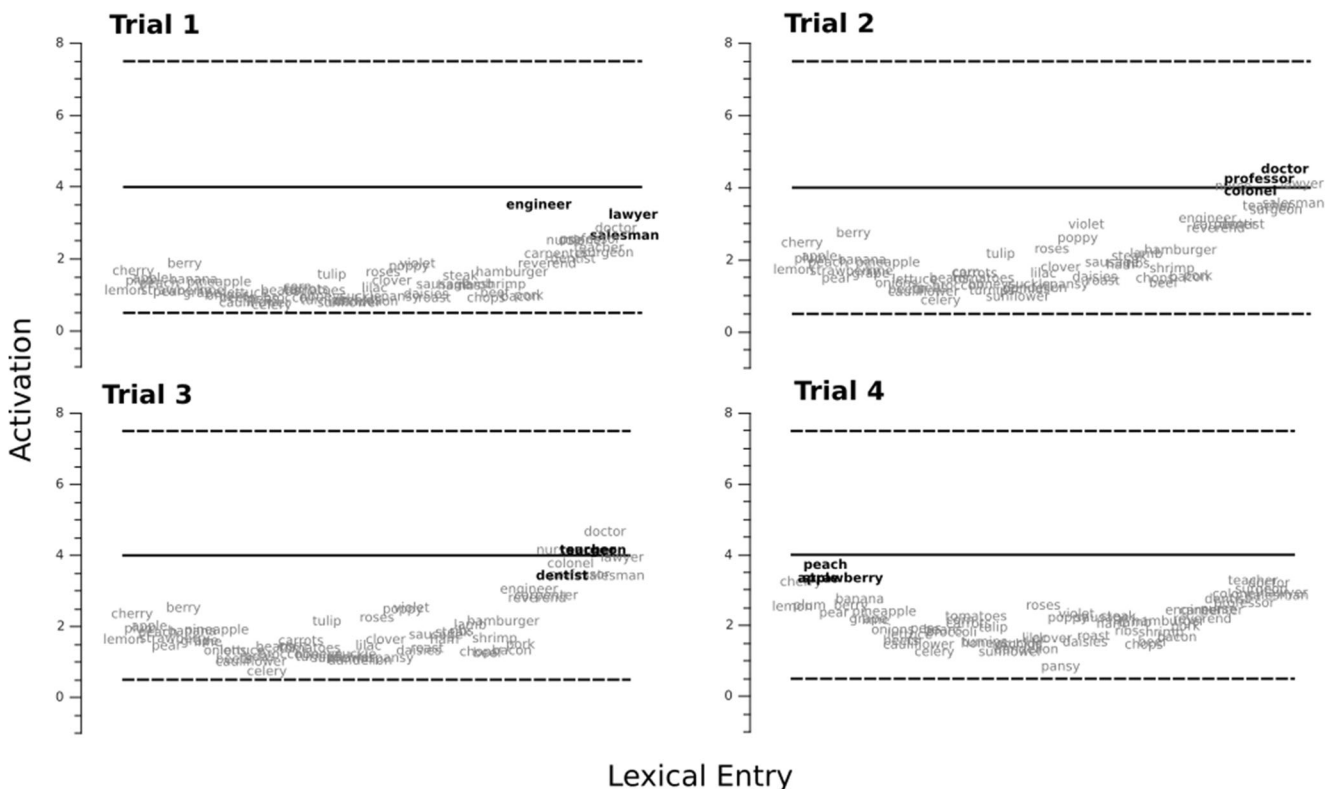


Fig. 6 Activation in the hologram in four stages. *Trial 1* shows activation after studying *engineer*, *lawyer*, and *salesman*. *Trial 2* and *Trial 3* show the corresponding data after studying *professor*, *doctor*, and *colonel*

followed by *teacher*, *dentist*, and *surgeon*, respectively. *Trial 4* shows the state of the hologram after studying *peach*, *strawberry*, and *apple*, words from a new category

respectively. The criterion was fixed at 1.0, with the range set to criterion ± 0.5 .

The lexical vectors used in the simulation, shown in Fig. 7, combine item and order information. Item and order vectors define different neighbourhoods (see M. N. Jones & Mewhort, 2007; Tables 1 and 2). To illustrate the difference, Table 3 shows the top 10 most similar items to the word *sail*, using

both the item and order vectors. Using item information, the words most similar to *sail* are all related to the sea, sailing, and travel. Using the order space, by contrast, the words are verbs. Because item vectors by themselves predict release from PI reasonably well (e.g., in the top panel of Fig. 3), the next simulations used item vectors rather than the lexical vectors used in Fig. 7.

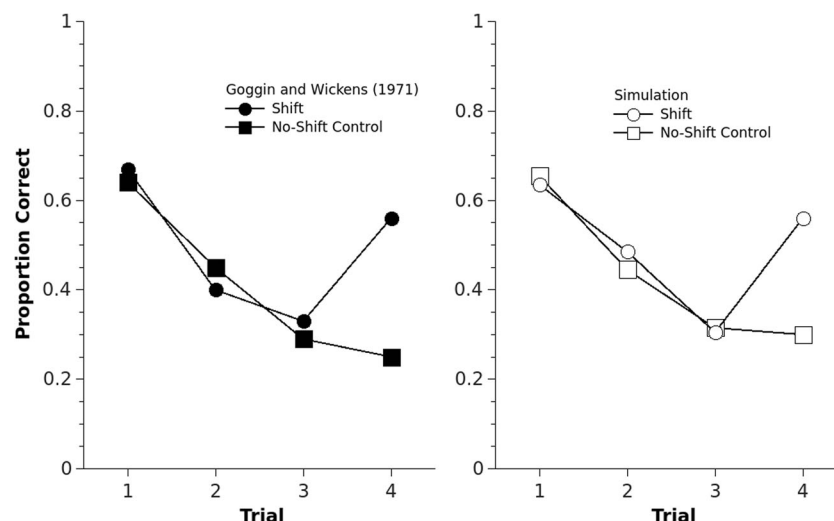


Fig. 7 Simulation of Goggin and Wickens (1971, Fig. 1) using lexical BEAGLE vectors

Table 2 Parameters for all fits

Experiment	Parameters						
	α	β	γ	λ	ω	range	n
Goggin and Wickens (1971)							
Food	0.470	0.587	0.717	-	-	0.50	60
Body	0.422	0.377	0.681	-	-	0.50	60
Wickens (1973)							
Fruits	0.214	0.681	0.431	-	-	0.50	40
Vegetables	0.217	0.773	0.454	-	-	0.50	40
Flowers	0.128	0.747	0.354	-	-	0.50	40
Meats	0.089	0.868	0.435	-	-	0.50	40
Professions	0.505	0.020	0.560	-	-	0.50	40
Keppel and Underwood (1962)	0.559	0.193	0.848	-	-	0.50	60
Loess and Waugh (1967)	0.180	0.706	0.415	-	-	0.50	40
Hebb (1961)	0.204	0.048	0.949	0.523	0.479	0.85	60
Von Restorff	0.506	0.819	0.776	0.862	0.730	0.50	200

In the Wickens (1973) study, shown in Fig. 1, the control condition did not include a switch of category (i.e., all trials were fruits); the other conditions switched from flowers to fruits, vegetables to fruits, meats to fruits, or professions to fruits.

Figure 8 shows our simulations of Wickens's data. Comparison of Fig. 1 and Fig. 8 indicates that the model provides a good account of release from PI in his task. Wickens (1970, 1973) discussed release from PI in terms of the overlap in features encoded for the various categories. For example, he distinguished meat from vegetables on the grounds that only one of the two is grown in the ground.

Table 3 Ten most similar words to sail in item and order vectors

Item		Order	
Word	Cosine	Word	Cosine
Sailing	.78	Cook	.87
Sails	.77	Fight	.86
Ship	.75	Drive	.85
Sailed	.73	Hunt	.85
Sea	.70	Report	.84
Ships	.70	Ride	.83
Vessel	.69	Work	.83
Sailors	.68	Blow	.83
Aboard	.66	Fly	.82
Boat	.66	Dance	.82
Pirates	.66	Swing	.82
Hull	.66	Swim	.82
Pirate	.66	Maneuver	.82
Anchor	.65	Cover	.81
Crew	.63	Transport	.81

The similarity structure of BEAGLE vectors, defined by usage in text, escapes the need to hand-craft an ad hoc rationale for differences amongst categories.

The von Restorff effect The release from PI examples, shown so far, reflect a change in meaning implemented by shifting one category to another. In a simple list-recall task, the von Restorff effect—an increase in accuracy when an element in a list is isolated—can be produced by using a list of words from

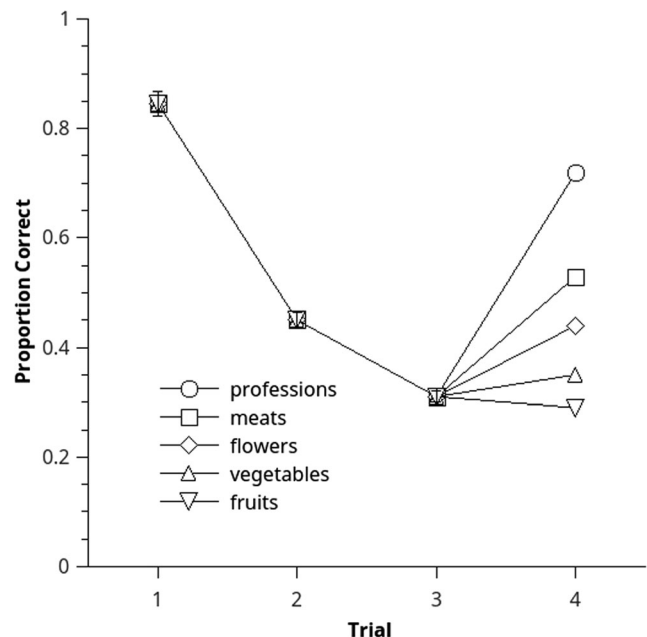


Fig. 8 Simulation for the fruits to fruits, vegetables to fruits, flowers to fruits, meats to fruits, and professions to fruits from Wickens, Dalezman, and Eggemeier (1976). The simulation used *item* vectors

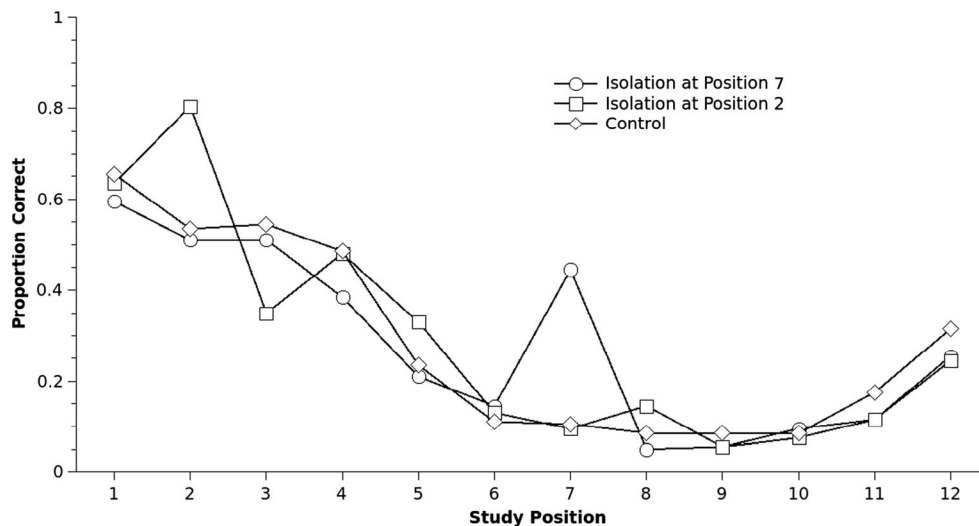


Fig. 9 Simulation of a von Restorff effect. *Open circles* show the data when the isolation word was in Position 7; *open squares* show the corresponding data when the isolation word was in Position 2. *Open diamonds* show the control condition

one semantic category, with the isolated item from a different semantic category.

To show that the mechanisms of the holographic model can capture the von Restorff effect, we implemented a recall task in which all but one of the words were from a single category (body parts); the single word was from a different category (foods). Each pseudosubject was provided 12 words; 11 of the words were selected at random from the set of 16 body parts; the isolated word was selected at random from the set of 16 foods. Because the study list was longer than the four-word lists used in the previous studies, we added serial-position weights for associative and item information. Specifically,

$$\mathbf{L} = \mathbf{L} + \alpha \times ((C - S_{\text{associative}}) + \lambda^{LL-i+1}) \times (\mathbf{w}_{i-1} \otimes \mathbf{w}_i) + \beta \times ((C - S_{\text{item}}) + \omega^i) \times \mathbf{w}_i,$$

where LL is the list-length, and i is the serial position from 1 to LL . As before, $S_{\text{associative}}$, and S_{item} are the current associative and item strengths.

Figure 9 shows the accuracy of report as a function of presentation position for both the control (no isolate) and the experimental (isolate) condition. The isolate (the *foods* word) was placed at positions seven two and seven. As is clear in Fig. 9, there was a robust benefit for the word in position seven, a semantic category advantage exemplifying the von Restorff effect.

An anonymous referee asked what the model predicts when the isolated word is the first or second in the list. The question was motivated by an attentional account for the von Restorff effect; the idea is that a subject will devote more processing to the unusual item. But an isolate advantage on the first or second position would rule out such an account because it is not until the third item that a subject would know which of the first three items is the unusual one. Hence, an

attentional strategy could not get started until the third item has been presented. The holographic model, of course, has no attentional mechanism and cannot be accused of devoting extra processing to a particular word.

To answer the referee's question, we moved the isolate to the second position. As shown in Figure 9, there was enhanced performance at Position 2.

Time-based proactive interference and release from PI

Keppel and Underwood Keppel and Underwood (1962) measured PI as a function of retention interval. Briefly, they reported that longer retention intervals led to greater PI. The holographic model explains both PI and release from PI in terms of changes to the hologram. Because the hologram changes only when information is added or subtracted, it is pertinent to ask what events could change the hologram in the Keppel and Underwood example.

The question of whether time can be a causal agent has a long history dating back to McGeoch's (1932) argument against memorial decay. Consistent with his view, we propose that an ostensibly unfilled interval is actually filled with random distractions and internally generated thoughts. Such filling alters the hologram by increasing its entropy. To implement the idea, we applied the model to Keppel and Underwood's task by adding a variable number of random Gaussian vectors to the hologram to simulate the interference associated with the passage of time. We have not formally scaled the number of added Gaussian vectors to time, but pilot data led us to weigh the Gaussian vectors by .38. Hence, we report the retention interval on an ordinal scale (i.e., *low*, *medium*, and *high*).

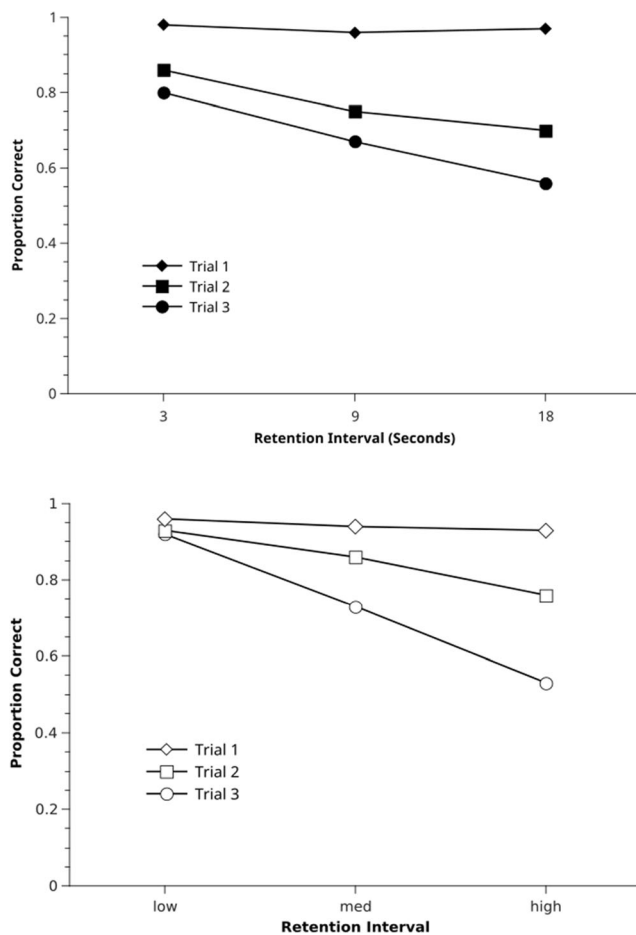


Fig. 10 Simulation of Keppel and Underwood (1962). *Diamonds* indicate performance on Trial 1; *squares* indicate performance on Trial 2; *circles* indicate performance on Trial 3

Figure 10 shows the effect of increasing retention interval on the severity of proactive interference. The top panel shows the data from Keppel and Underwood (1962, Experiment 2). The bottom panel shows the simulation. Comparison of the two panels shows that the simulation matches the original data remarkably well.

Brown, Neath, and Chater All the examples of release from PI so far, have been examples of semantic contrast. Other procedures produce release by placing a rest interval between the last and second-last trials, without a shift in the category of words (e.g., Loess & Waugh, 1967). Temporally induced release is usually discussed in terms of a distinctiveness construct rather than in terms of the semantics of item representation (e.g., Brown, Neath, & Chater, 2007).

At first blush, it may be tempting to treat temporal release as a different phenomenon, one unrelated to semantic release. Although that possibility remains open to empirical scrutiny, Brown et al. (2007) argue that time-based release reflects item discriminability (distinctiveness). In their model, SIMPLE, they treat time on a psychological dimension (a logarithmic

scale). Consequently, release from PI falls out of the temporal discriminability of the items on the scale. If one assumes an analogous scale for semantic distinctiveness, they argue that, in principle, the same distinctiveness idea could explain semantic release as well as time-based release. In light of their argument that semantic and temporal release can be brought under the same theoretical umbrella, we applied the hologram model to their example.

As we argued in connection with Keppel and Underwood's (1962) study, changes to the hologram occur only when items are added to, or subtracted from, it. Hence, we added random Gaussian vectors to simulate random events during the passage of time. Likewise, to simulate the passage of time during Brown et al.'s temporal interval, we added random Gaussian vectors. Although we added Gaussian vectors in both cases, the effect of time is different in the two; Keppel and Underwood report interference whereas Brown et al. (2007) report release from interference.

For the paradigm discussed by Brown et al. (2007; Loess & Waugh, 1967), the temporal manipulation is not a retention interval (as in Keppel & Underwood's, 1962, paradigm) but a rest interval between the penultimate trial and the final trial. Although the time interval may be the same in both cases, from a subject's perspective, the task demands are quite different. When time is introduced in a retention interval, the subject must work hard to fight against the increasing entropy that develops to maintain the integrity of the studied words. When a comparable interval is introduced as a rest, by contrast, rather than maintaining previously studied and reported information, the subject is free to reduce memory load; that is, to reset memory to a neutral state.

Because the task demands are different for rest and retention, when we implemented the rest-interval paradigm, we filled the time interval with Gaussian vectors, as before, but to acknowledge the difference in task demands, before the final trial was administered, we normalized the hologram to reset it.

Figure 11 shows release from PI using content-free interference to force the release. The top panel of Fig. 11 shows items recalled correctly as a function of trials and a temporal interval between the third and fourth learning trials. As is shown in the top panel of Fig. 11, the amount of release increased as the number of content-free vectors was added to the hologram. We used the same category of materials on all trials; therefore, all changes reflect the passage of time.

The bottom panel of Fig. 11 shows the size of the release (using Wickens's, 1970, measure) as a function of the number of content-free vectors. As shown, the size of the release is an increasing exponential function of the number of content-free additions, our surrogate of retention interval.

The form of the function shown in the bottom panel of Fig. 11 resonates nicely with Brown et al.'s (2007) arguments about distinctiveness in the SIMPLE model. The holographic

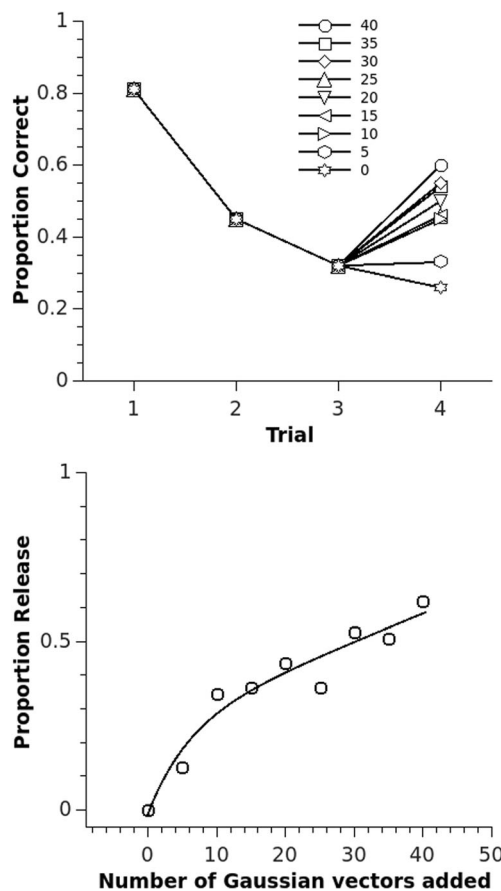


Fig. 11 *Top panel* shows simulated RPI as a function of trials and the number of content-free interference vectors added to the hologram. *Bottom panel* shows the size of the RPI as a function of the number of content-free interference vectors added. The increasing data are fitted by an increasing exponential growth function

model may provide a mechanism by which distinctiveness can be understood.

Related paradigms

Release from PI reflecting contextual overlap Lohnas, Polyn, and Kahana (2015; see also Howard & Kahana, 2002) have shown that release from PI can result from an interaction between episodic (temporal) context and semantic context reflecting pre-experimental associations shared by words within the same category. They define context as “the set of features surrounding but not comprising the memory itself” (p. 337), and they use orthonormal basis vectors (i.e., vectors of zeroes with a single element set to one) to identify individual words. Basis vectors identify a word in the sense that each vector is unique, but they do not contain information about the word’s meaning or the words’ relation to other words. Their model uses patterns of random Gaussian vectors

to represent a word’s episodic (temporal) context, and because basis vectors do not contain information about meaning, all of the work in their model is based on changes to the context.

Recall is driven by an interaction between the item and context information: A probe item reinstates the context associated with that item during study; the reinstated context, in turn, updates the item information associated with it, leading to the retrieval of a studied item. The retrieved item becomes a cue in preparation for report of the next item. In this way, multiple-word recall is driven by a series of context-to-item and item-to-context cycles.

On the assumption that semantically related items share context, the preexperimental context-to-item associations are initialized, in their simulation, so that items have high contextual overlap within a category but low contextual overlap across categories. PI occurs because semantically related items from prior study lists overlap contextually with items from the immediately studied list. Release from PI occurs because conflict between semantic and temporal context is attenuated when a new category’s context is presented.

Because the semantic context reflects preexperimental associations, Lohnas et al.’s (2015) demonstration of release from PI has been handcrafted on exactly the question of the semantic-context association. It is an open question whether one can construct semantic context-to-item associations with sufficient resolution to mimic the structure learned by the BEAGLE algorithm. But that is what would be needed to match the word-specific release effects illustrated here (but see Howard, Shankar, & Jagadison, 2010; Rao & Howard, 2008).

Lohnas et al. (2015) place responsibility for PI effects on context, whereas our account places it on item representation and changes in representation strength afforded by the hologram. Both accounts are incomplete: Lohnas et al. would benefit by including a more complete item representation, and the present account would benefit by adding a mechanism to support list discrimination and like context effects.

The Hebb repeated-list effect In a classic chapter, Hebb (1961) distinguished an initial short-term memory trace based on “reverberatory [neural] activity set up without necessarily depending on any change in the units involved” from a longer-term trace that “consists of some change in the units which outlasts their period of activity” (p. 41). He called the first kind of trace an *activity* trace and the second a *structural* trace. To test whether information in an activity trace can survive competition from successive trials, he administered a series of memory-span trials. On each trial, the subject listened to a series of nine digits and then recalled the digits in the order in which they had been presented. Unknown to the subjects, every third trial repeated the same sequence heard on the first trial. Performance on the nonrepeated trials was stable, but, to Hebb’s surprise, performance on the repeated trials increased

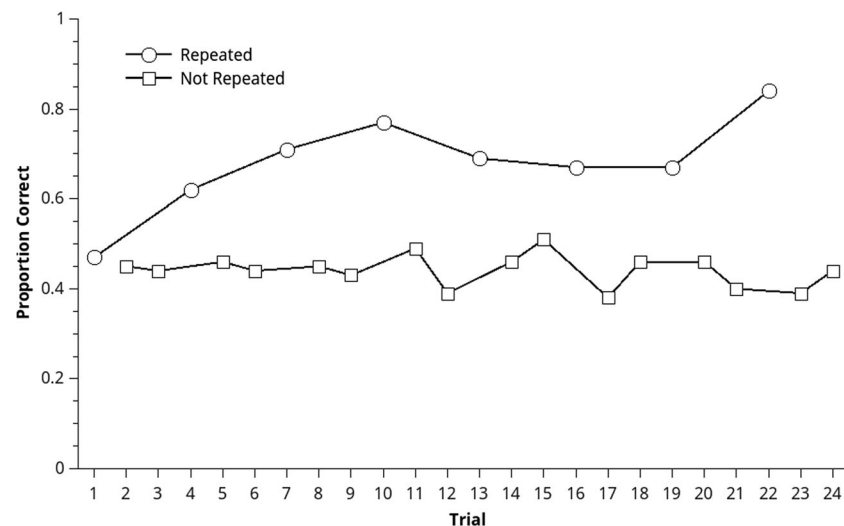


Fig. 12 Simulation of the Hebb repeated list task

relative to the nonrepeated trials. Clearly, information from the repeated activity traces persisted even though it should have been overwritten by traces of the trials intervening between the repeated series. In terms of the activity-structural dichotomy, repeated trials must have induced a structural change.

The activity-structural dichotomy has no parallel in the holographic model. Activation can be pushed up or down by events on a trial or series of trials. The question, then, is whether the model can also capture the repeated-list paradigm.

To answer the question, we implemented a series of 24 trials using nine words from various categories at random. Every third trial (i.e., trial 1, 4, 7 . . .) was a repeat of the list used on the first trial. Each pseudosubject received a permutation of the words.

Figure 12 shows the results of the simulation. As is clear in Fig. 12, the accuracy on the nonrepeated trials was stable across. Accuracy on the repeated trials, by contrast, increased slowly as the number of repetitions increased.

Hebb (1961) undertook the repeated-list task to test persistence of memory. His original idea assumed a labile short-term memory, an idea popular at the time (see G. Jones & Macken, 2015, for a discussion). Holographic storage is permanent but the strength of representation for individual items and associations can rise and fall. The hologram yields the repeated-list effect because information—particularly about inter-item associations—persists and accumulates.

Discussion

In a classic paper, Wickens (1970) showed that the size of the release from PI depends on the semantic categories used first to induce PI and then to recover from PI. Some pairs of categories yield more release than others.

Why do different combinations of categories induce different degrees of release from PI? We have shown that the differences reflect the degree of representational overlap between the pairs of categories. Specifically, we examined the BEAGLE vectors corresponding to the words used by Wickens (1970) and showed that the size of release is a function of the Euclidean distance between the inducing and the release categories (weighted by a measure of the way the words cluster around the centre of each category).

To illustrate how the information in the BEAGLE vectors can be used in immediate recall experiments, we turned to Franklin and Mewhort's (2015) holographic account of recall and equipped its lexicon with BEAGLE vectors corresponding to the words that Wickens (1970) and his colleagues used in their studies. The model captured examples of release from PI, thereby confirming that BEAGLE vectors represent meaning in a computationally tractable way.

The idea of using a hologram as a storage and processing mechanism is novel. Hence, a comment is in order about how it fits (or does not fit) with several concepts popular in the literature:

1. The pool of words (39,076 in the present corpus) represents all possible words that a subject might know. We built a hologram for each pseudosubject by assigning 10,000 words (all but the studied material at random). By sampling from the pool of possible words, we were able to create artificial subjects and to facilitate fitting by averaging across them. The practice also acknowledges that subjects are unlikely to know all of the same words.

We used the hologram to hold a pseudosubject's lexicon. One's lexicon is usually considered an example of a long-term memory. For that reason, conventional treatment anticipates

that it will be a static store.³ But, as illustrated here, it is anything but static. The hologram is an example of permanent yet dynamic storage. Its dynamic property allows it to serve as a basic processing mechanism as well as a storage system; hence, it can participate in categorization.

Dynamic data structures are often distinguished from static ones on the basis of their extendibility. A static array has a fixed number of elements whereas a dynamic array can extend the number of elements. We use the term dynamic array differently: The hologram is a dynamic array in the sense that it encompasses multiple static arrays (BEAGLE vectors) so that the strength of representation of each can be altered separately. In the holographic account, the 10,000 words used to form the lexicon are static, whereas the hologram itself—a single vector—is dynamic. In a sense, our framework agrees with the standard view that the memory system contains both static and dynamic representations. Of course, the decision to assign 10,000 words to a lexicon was arbitrary, as was the decision to use a pool of 39,076 words.

2. Tulving (1972, 1985) has popularized the distinction between episodic and semantic memory. His distinction is well accepted in current work. Nevertheless, it is readily apparent that meaning matters in performance of episodic experiments. To acknowledge meaning, people frequently appeal to top-down-processing, typically without offering a mechanism to implement it. Except for examples of deep learning in neural nets (LeCun, Bengio, & Hinton, 2015), few have proposed that the system combines episodic and semantic information (but see Elias Smith, 2013). The hologram offers a concrete computationally tractable mechanism to implement top-down processing in an episodic memory task. In effect, it provides an interface between semantic and episodic memory.

As we noted earlier, however, the holographic account is incomplete because it does not currently retain context information. A holographic model that defines and stores both context and content information is currently under development.

3. The memory literature has enjoyed a long, and sometimes heated, debate about decay versus interference as the cause of loss from memory (e.g., Farrell et al., 2016; Lewandowsky, Oberauer, & Brown, 2015). The hologram is a permanent memory, but, because the level of activation for particular items can fall when unrelated material is studied, it may mimic a decay process. Critically,

however, all changes in the hologram are event driven; that is, the passage of time does not, by itself, affect activation. Hence, holographic model falls into the camp of interference theories.

4. The model requires a retrieval cue to prompt recall. Traditionally, a retrieval cue has been thought to work because it is linked, somehow, to the item to be recalled; that is, the cue is part of an associative mechanism. In the holographic model, however, a retrieval cue does more than point to an associated item: encoding a retrieval cue alters the strength of representation of all items in the hologram.

Final thoughts The present work is intended to focus theory on content rather than on context, but the important question is how content and context information combine. As we have illustrated, meaning drives recall. Introspection suggests that the meaning of one thought determines, at least in part, the thought to follow. The challenge, then, is to specify how all information available to the system is combined to determine how one thought follows another.

Finally, recall that Wickens (1970) viewed encoding as a complex operation. Encoding meant that a word as a token was brought to the memory system where its meaning was filtered to the appropriate categories. *Horse* is not a simple token but a pointer to concepts related to it: It points to a rich list of binary taxonomies “in the manner of a good player of Twenty Questions, but in more or less the reverse direction” (Wickens, 1970, p. 1).

Wickens (1970) did not specify the source of the knowledge underlying the taxonomy. Paradoxically, simply specifying a taxonomy is not enough; it is easy to define a taxonomy based on the distinction between verbs and adjectives, but as Wickens showed, that distinction does not induce release from PI. Therefore, a taxonomic distinction is a necessary, but not sufficient, specification.

BEAGLE’s success in accommodating release from PI provides a hint about the source of knowledge. In Fig. 6, when words were encoded, their level of activation increased. Importantly, activation of related words that were not encoded also increased. For example, encoding *engineer*, *lawyer*, and *salesman* not only increased the activation of those words but also increased the activation of *doctor*, a related word but not a studied word. The relation is not associative—there is no reason to suspect that words within a category have been deliberately learned as an associative bundle. Rather, the relation falls out of the similarity structure of the vectors, and the similarity structure is a function of knowledge of language. The BEAGLE vector for a word is a history of the ways in which the word has been used in the sample of language (the corpus) determining the construction of the vector. The pattern of usage in the language itself is, at least in part, the source of the knowledge used when encoding a word.

³ Shiffrin (1999, p. 20) suggested that “the primary structural distinction in the memory system [of the Atkinson & Shiffrin, 1968, model] is between the active memories (all the short-term stores and sensory stores) and the passive memory (long-term store).

Compliance with ethical standards

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Appendix

Words used to calculate overlap

Baldwin and Wickens (1974)

Number of syllables (NS)

1 Syllable: bread Maine nose gun bee sale house nurse fox green train rose saw mumps flute woods nun yacht boat hill pearl stove prose dime lake chair wool deer tea church elm skirt crow soap trout pen beer golf mile storm pea hour aunt door gold France school pear

2 Syllables: butter Texas finger rifle spider pepper hotel doctor rabbit yellow airplane daisy hammer measles trombone forest pastor sailboat wagon valley ruby furnace poem nickel river table cotton lion coffee temple maple sweater eagle towel salmon pencil whiskey tennis meter lightning carrot minute uncle window silver England college apple

Number of phonemes (NP)

Two or three phonemes: dumb hay eel log toe day edge fog ear odd debt ash

Four or five phonemes: smart field shark branch wrist month fringe breeze throat freak thrift slag

Acoustic & articulation (AA)

Closed front: pig lid hip niche fierce east sea wheat inch flea ditch sleet skiff disc twig sheep yeast wind eel sleeve bee dish splint finch

Open back: hog cork jaw slot harsh north shore corn yard wasp gorge fog yacht orb straw fox lard storm cod smock moth fork gauze swan

Swanson and Wickens (1970)

Frequency of nouns (Fr)

High-frequency nouns: nature chair home music ball office room nation truth valley glass snow water inch doctor cotton party life night book artist wall method judge table garden fact gate river minute

Low-frequency nouns: turnip niche jade chalk herb puppet soot attic instep cavern idol posse helm banjo rodent racket pedal knob lobe cactus gull nectar logic mutton pact index veal mantel

Frequency of verbs (Fr)

High-frequency verbs: notice think join tell accept serve begin give decide play finish count drive send arrive obtain open learn save change study teach vote settle expect laugh keep know write forget

Low-frequency verbs: amend veer bisect exhale broil romp omit vacate yearn prance munch cede chew delude affix tangle gulp defer exert bounce lure oust wring invert misuse pave nudge can ponder lurch

Reutener (1972)

Words versus numbers (WN)

Words: chair table bed sofa desk lamp couch dresser television stool rug bookcase cabinet chest piano footstool buffet bench

Numbers: one two three four five six seven eight nine ten eleven twelve

Bousfield, Cohen, and Whitmarsh (1958)

Taxonomic category (TC)

High word frequency

Animals: dog cat horse cow bear lion deer fox rabbit tiger

Names: John Bob Joe Bill Jim Tom Richard George Jack Frank

Countries: France England Germany Russia Spain Canada Italy Mexico Japan Sweden

Cloths: cotton wool silk rayon nylon linen satin velvet

Vegetables: carrot pea potato bean corn lettuce spinach squash cabbage turnip professions

Professions: doctor lawyer professor dentist teacher reverend nurse engineer carpenter salesmen

Musical instruments: piano violin trumpet clarinet flute trombone oboe harp guitar cello

Birds: robin sparrow blue jay canary crow wren eagle oriole hawk parrot

Low word frequency

Animals: wolf squirrel leopard raccoon donkey zebra beaver kangaroo buffalo camel

Names: Michael Arthur Carl Stephen Ralph Philip Walter Bernard Dennis

Countries: Scotland Poland Egypt Holland Belgium Finland Greece Turkey Denmark Cuba

Cloths: flannel muslin gingham corduroy burlap denim calico poplin khaki

Vegetables: radish cauliflower cucumber pepper parsnip pumpkin mushroom artichoke melon yam

Professions: accountant farmer banker chemist plumber mechanic painter butcher grocer janitor

Musical instruments: bugle banjo bassoon piccolo cornet harmonica lute mandolin

Birds: pigeon owl lark dove ostrich falcon peacock stork penguin

Wickens, Reutener, and Eggemeier (1972)

Sensory Impression (SI)

Round: barrel knob doughnut head balloon wheel globe baseball spool dome saucer button

White: milk teeth napkin linen rice bandage chalk lint salt snow collar diaper

Wickens and Clark (1968)

Semantic differential (SD)

Evaluation (SD_E)

High: church god beauty pleasant hand home love fine music religious friend peace marry friendly rest

Low: war fire disease bad enemy kill terrible hate debt failure battle argue fight death danger

Activity (SD_A)

High: fire sailor great attack war win party love bird arms star laugh kiss arm warn

Low: dead silent sleep lie rock egg still stone silence window quiet rest die late iron

Potency (SD_P)

High: steel iron rock metal stone hard building law strong tree oil mountain power science duty

Low: love kiss baby wife girl beautiful sister mother woman heart young music sorrow daughter child

Wickens and Engle (1970)

Imagery (Im)

Low: surtax functionary criterion inanity impropriety concept interim gist instance kine banality abasement unreality debacle soul allegory context disclosure encephalon idea fact tendency foible hankering equity exactitude discretion aberration origin figment

High: coffee sea hammer frog library garden ocean mountain yacht leopard ankle tree orchestra breast bouquet brassiere horse cat strawberry kiss cigar lemon fireplace sunset elephant alligator car automobile girl snake

Wickens, Clark, Hill, and Wittlinger (1968)

Verbs versus adjectives (VA)

Verbs: accept add admire admit advise afford agree alert allow amuse announce annoy answer appear applaud appreciate approve argue arrange arrest arrive ask attach attack attempt attend attract avoid back bake balance ban bang bare bat bathe battle beam beg behave belong bleach bless blind blink blot blush boast boil bolt bomb book bore borrow bounce bow box brake branch breathe bruise brush bubble bump burn bury buzz calculate call camp care carry carve challenge change charge chase cheat check cheer chew choke chop claim clap clean clear clip close coach coil collect comb command communicate compare compete complain complete concentrate concern confess confuse connect consider consist contain continue copy correct cough count cover crack crash crawl cross crush cry cure curl curve cycle dam damage dance dare decay deceive decide decorate delay delight deliver depend describe desert deserve destroy detect develop disagree disappear disapprove disarm discover dislike divide double doubt drag drain dream dress drip drop drown drum dry dust earn educate embarrass employ empty encourage end enjoy enter entertain escape examine excite excuse exercise exist expand expect explain explode extend face fade fail fancy fasten fear fence fetch file fill film fire fit fix flap flash float flood flow flower fold follow fool force form found frame frighten fry gather gaze glow glue grab grate grease greet grin grip groan guarantee guard guess guide hammer hand handle hang happen

harass harm hate haunt head heal heap heat help hook hop hope hover hug hum hunt hurry identify ignore imagine impress improve include increase influence inform inject injure instruct intend interest interfere interrupt introduce invent invite irritate itch jail jam jog join joke judge juggle jump kick kill kiss kneel knit knock knot label land last laugh launch learn level license lick lie lighten like list listen live load lock long look love man manage march mark marry match mate matter measure meddle melt mend mess up milk mine miss mix moan moor mourn move muddle mug multiply murder nail name need nest nod note notice number obey object observe obtain occur offend offer open order overflow owe own pack paddle paint park part pass paste pat pause peck

Adjectives: adorable beautiful clean drab elegant fancy glamorous handsome long magnificent plain quaint sparkling unsightly red orange yellow green blue purple gray black white alive better careful clever dead easy famous gifted helpful important inexpensive mushy odd powerful rich shy tender uninterested vast wrong angry bewildered clumsy defeated embarrassed fierce grumpy helpless itchy jealous lazy mysterious nervous obnoxious panicky repulsive scary thoughtless uptight worried agreeable brave calm delightful eager faithful gentle happy jolly kind lively nice obedient proud relieved silly thankful victorious witty zealous broad chubby crooked curved deep flat high hollow low narrow round shallow skinny square steep straight wide big colossal fat gigantic great huge immense large little mammoth massive miniature petite puny scrawny short small tall teeny tiny cooing faint hissing loud melodic noisy purring quiet screeching thundering voiceless whispering ancient brief early fast late long modern old quick rapid short slow swift young bitter delicious fresh greasy juicy hot icy loose melted nutritious prickly rainy rotten salty sticky strong sweet tart tasteless uneven weak wet wooden boiling breeze broken bumpy chilly cold cool creepy crooked curly damaged damp dirty dry dusty filthy flaky fluffy freezing hot warm wet abundant empty few full heavy light many numerous sparse substantial

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